

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

02

Q1

GRID-IN ALGEBRA

What is the slope of the graph of $y = \frac{1}{4}27x + 15 + 7x$ in the xy -plane?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN

ALGEBRA

If $2(3t - 10) + t = 40 + 4t$, what is the value of $3t$?

STUDENT-PRODUCED RESPONSE

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.	/	.	/	.
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0	0	0	0
---	---	---	---

1	1	1	1
---	---	---	---

2	2	2	2
---	---	---	---

3	3	3	3
---	---	---	---

4	4	4	4
---	---	---	---

5	5	5	5
---	---	---	---

6	6	6	6
---	---	---	---

7	7	7	7
---	---	---	---

8	8	8	8
---	---	---	---

9	9	9	9
---	---	---	---

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In the linear function f , $f(0) = 8$ and $f(1) = 12$. Which equation defines f ?

A. $f(x) = 12x + 8$

B. $f(x) = 4x$

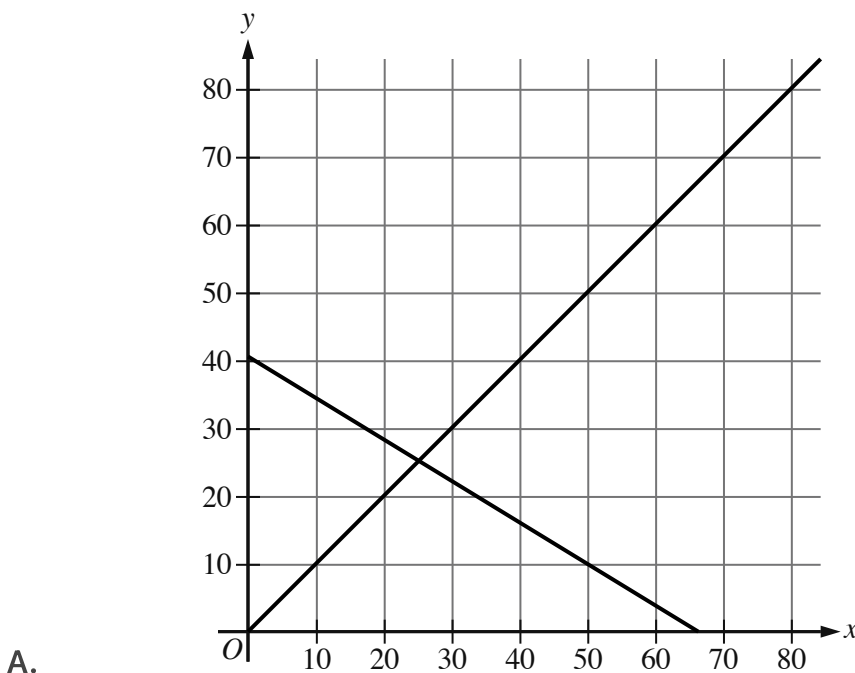
C. $f(x) = 4x + 12$

D. $f(x) = 4x + 8$

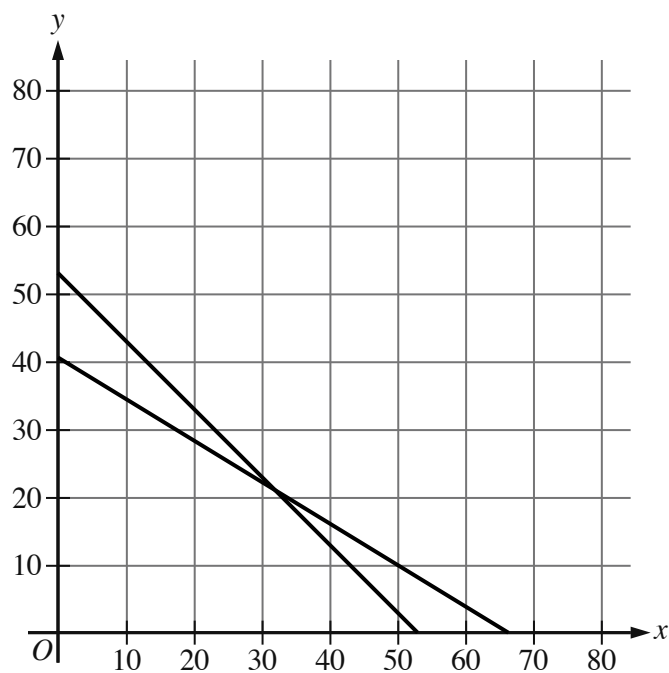
$$x + y = 53$$

$$11x + 18y = 730$$

The given equations represent the possible numbers of beach chairs, x , and umbrellas, y , rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?



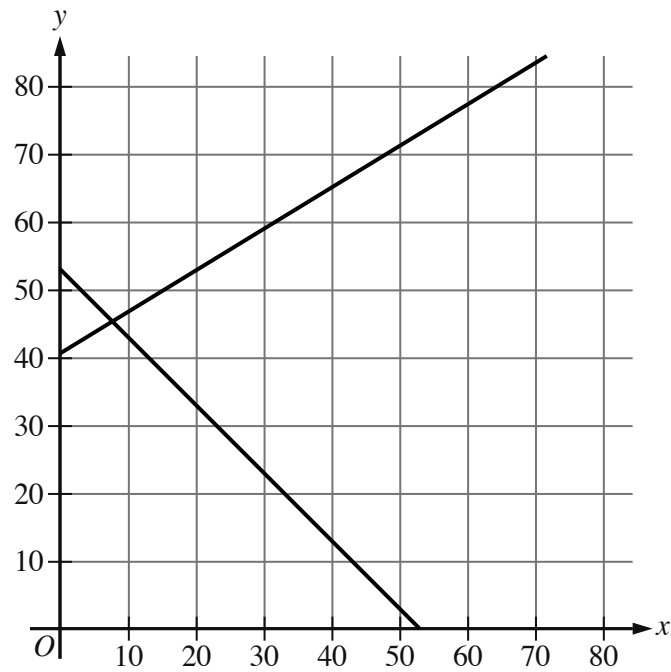
- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (66 comma 0)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma 0)
 - (50 comma 50)
- The 2 lines intersect at 1 point.



B.

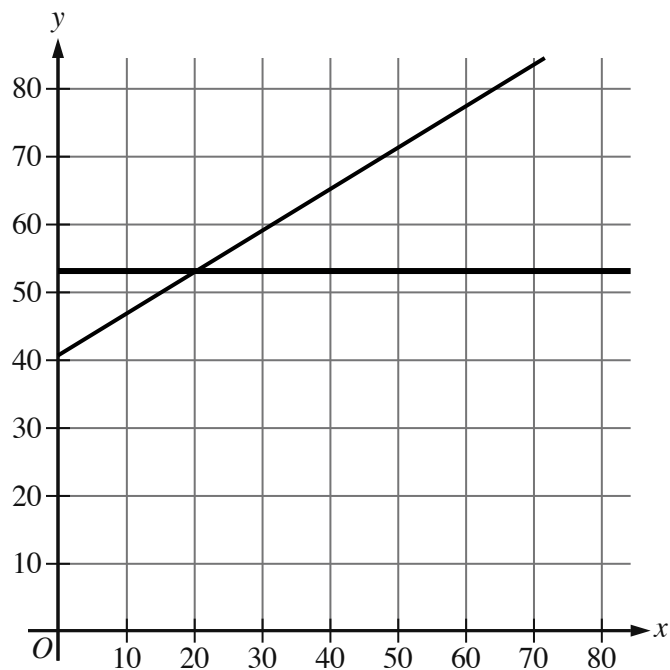
- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 53)
 - (53 comma 0)
- For the second line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (66 comma 0)
- The 2 lines intersect at 1 point.

C.

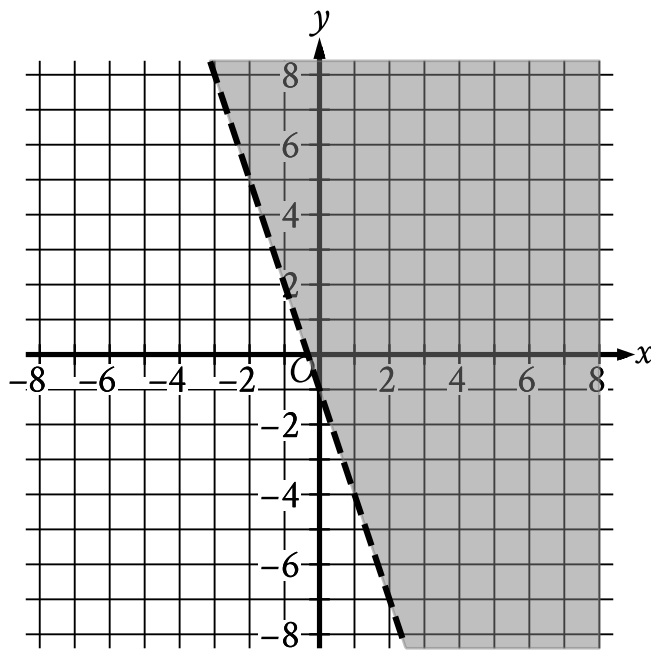


- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 53)
 - (53 comma 0)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (30 comma 59)
- The 2 lines intersect at 1 point.

D.



- For the first line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (0 comma 53)
 - (30 comma 53)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (30 comma 59)
- The 2 lines intersect at 1 point.



- The boundary of the inequality is a dashed line.
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (negative 1 comma 2)
 - (0 comma negative 1)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

- A. $y < -1 + 3x$
- B. $y < -1 - 3x$
- C. $y > -1 + 3x$
- D. $y > -1 - 3x$

Q6**ALGEBRA**

What is the slope of the graph of $10x - 5y = -12$ in the xy -plane?

A. -2

B. $-\frac{5}{6}$

C. $\frac{5}{6}$

D. 2

Q7**ALGEBRA**

$$\frac{1}{4}x + 5 - \frac{1}{3}x + 5 = -7$$

What value of x is the solution to the given equation?

A. -12

B. -5

C. 79

D. 204

$$h(x) = x + b$$

For the linear function h , b is a constant and $h(0) = 45$. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

$$y = 3x$$

$$2x + y = 12$$

The solution to the given system of equations is x, y . What is the value of $5x$?

- A. 24
- B. 15
- C. 12
- D. 5

Q10

ALGEBRA

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h , in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights h , in centimeters, for a tall plant species?

- A. $106 < h < 158$
- B. $80 < h < 100$
- C. $42 < h < 87$
- D. $17 < h < 85$